# SGDAgent Reasoning Loop

**Question:** How many white lights were on at 45:08?

**Agent (Step 1):**
Calling tool **Frame Inspect** at ["00:45:08", "00:45:10"] with query: "How many white lights are on?"

**Frame Inspect:**
Fine-grained Response:
There are **34** white lights on. (✗ **Wrong**)

**Agent (Step 2):**
Calling tool **Global-Level Retrieval** with query: "white lights".

**Global-Level Retrieval:**
Matched Events:
evt_0053 [44:35–45:30] score=2.6053, confidence=0.5975
stage, lights; matched participant: **ent_000023 = lights** (confidence=0.95)
.....
evt_0022 [16:40–17:10] darkness, lights
evt_0056 [47:30–48:00] lights, screens, stage
.....

**Agent (Step 3):**
Calling tool **Segment-Level Retrieval** with query "white lights".

**Segment-Level Retrieval :**
Matched Segments:
seg_0108 [44:35–45:05] score=2.4863; entities: stage, white lights, colored lights; predicates: none; matched entity: **ent_000026 = white lights** (confidence=0.95)
.....
seg_0109 [45:00–45:30] score=2.4863; entities: stage, lights; predicates: none
seg_0184 [75:15–76:40] score=2.4316; entities: person, stage, crowd, lights
.....

**Agent (Step 4):**
Calling tool **Frame Inspect** at ["00:45:08", "00:45:09"]:
"Count individual white lights and disregard colored lights."

**Frame Inspect:**
Fine-grained Response:
The images show **24** individual white lights.
**10 left + 10 right + 4 center**

## Hierarchical Spatio-Temporal Scene Graph Representation

